

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.E. / B.Tech I Sem (Main) Examination March 2023

Problem Solving and Programming

(Common to all branches)

Time: 3 Hours

Max Marks: 60

Note: Answer ALL questions from Part-A at one place in the same order and Part-B (Internal Choice)

Part - A

(5Q X 2M = 10 Marks)

	M	CO	BT
1 Design a flowchart for doing sum of two numbers.	(2)	4	3
2 Compare <i>lists</i> and <i>tuples</i> with appropriate examples.	(2)	3	4
3 Differentiate between sequence mutation and accumulation patterns.	(2)	1	2
4 List any two advantages of modularizing program into functions.	(2)	3	1
5 What is a file? List any two file related functions/methods.	(2)	6	1

Part - B (50 Marks)

(5Q X 10M = 50 Marks)

	M	CO	BT
6 (a) Summarize evolution of programming languages with appropriate examples.	(5)	1	2
(b) What is an algorithm? Develop an algorithm for finding greatest of three numbers?	(5)	4	3
(OR)			
7 (a) Explain program development steps.	(5)	1	2
(b) Discuss software requirements for programming.	(5)	1	2
8 (a) What is a String? Explain any five string handling functions with suitable examples.	(5)	3	2
(b) Categorize the scope and lifetime of local and global variables in Python.	(5)	3	1
(OR)			
9 (a) Explain about <i>arithmetic</i> and <i>relational</i> operators with appropriate examples.	(5)	2	2
(b) Develop a Python program to generate prime numbers in a given range.	(5)	2	3
10 (a) Describe iterative statements in Python with suitable program.	(5)	2	2
(b) Write a Python program to find factorial of a given number.	(5)	3	3
(OR)			
11 (a) Explain the construct of <i>nested if-else</i> statement with a suitable example.	(5)	2	2
(b) Develop a python program to do matrix addition.	(5)	3	3
12 (a) Develop a Python program to read a word and print the number of letters, vowels and percentage of vowels in the word using a dictionary.	(5)	5	3
(b) Develop a Python program to generate Fibonacci series using a recursive function.	(5)	3	3

(OR)

- 13 (a) What is recursion? Develop a Python script to find the Greatest Common Divisor (GCD) of given two numbers using recursion. (5) 3 1,3
(b) Explain various dictionary methods with suitable examples. (5) 5 2
- 14 (a) How to create a text file and write content in to the text file? Explain with a python script. (5) 6 2
(b) Explain about Python memory management operations. (5) 6 2
- (OR)**
- 15 (a) Develop a python program that reads the characters from a file and displays those characters on the screen in capital form.
(Example the file name is a '*sample.txt*' and its contents are "*Cbit Hyd*", then the output should be "*CBIT HYD*") (5) 6 3
(b) Explain different modes to open a file with suitable examples. (5) 6 2
